# ECON-7102
# Environmental Economics I*

Prof. Casey J. Wichman

Georgia Institute of Technology

Fall 2026

**Course website:** `https://caseyjwichman.com/econ7102/`
**Course meeting time:** Mondays and Wednesdays, 11:00 AM – 12:15 PM
**Classroom:** West Architecture 259
**Office:** Old Civil Engineering 310
**Office hours:** By appointment

**Course description:** This is the first course in the Environmental Economics Ph.D. sequence. In this course, we will cover the theory of externalities and public goods related to the environment, followed by an exploration of policy instruments designed to address environmental problems. Next, we will revisit concepts of economic welfare and the tools economists use to value non-market environmental goods. We will also cover models that deal with the management of renewable and non-renewable natural resources. The final portion of the course will survey relevant topics in contemporary environmental economics to expose students to the research frontier.

*Prof. Casey Wichman: wichman@gatech.edu, School of Economics, Georgia Institute of Technology, 221 Bobby Dodd Way, Atlanta, GA 30332. **Last update:** August 20, 2026.

# 1 Course Overview

## 1.1 Course objectives

This course is intended to provide exposure to the theoretical foundations of environmental economics—particularly with respect to policy instrument choice and non-market valuation techniques—and give students a suite of tools to engage with the academic literature in environmental and resource economics. In addition, students will be exposed to the frontier of knowledge in the field and will develop a critical skillset to evaluate that research. Finally, students will take a step forward in identifying and pursuing a research question of their own interest that will, ideally, form the basis of a successful dissertation chapter or journal article.

## 1.2 Course goals

This course has three goals. First, students will learn fundamental concepts that underlie environmental and resource economics from a theoretical perspective. The first section of the course will cover the causes and consequences of environmental externalities and how to model potential solutions. Then, we will cover a variety of tools used to value environmental goods, followed by natural resource management issues. Second, students will be exposed to a variety of research at the frontier of the field, ranging from environmental justice to estimating climate change impacts, from consumer responses to electricity pricing to valuing health impacts of air pollution. Third, students will be expected to take a step towards producing, rather than consuming, research. In that effort, a subtext of this course will be in developing the craft of research, primarily focusing on how to identify and execute tangible research ideas. The second course in this sequence (ECON 7103 taught by Prof. Dylan Brewer) will focus more strongly on putting those ideas into action empirically.

# 2 Course Expectations

## 2.1 Attendance

Attendance is expected and will be necessary to succeed in the course. Participation in class discussions will be strongly encouraged. Ask questions. If you do not understand something in class, you are almost certainly not the only one. If you never volunteer an incorrect answer, you aren't taking enough risks.

## 2.2 Course website and Canvas

The course schedule, slides, and journal-article readings live on the course website: `https://caseyjwichman.com/econ7102/`. The website is the current version of the schedule and reading list and will be updated as the semester goes on.

Canvas is used for assignment submissions, grades, announcements, and reference materials such as textbook PDFs.

### 2.3 Grading

The final grade is calculated as follows:

| | |
|---|---|
| "Weekly" reading responses | 20 pts |
| Research paper proposal | 35 pts |
| Midterm | 25 pts |
| Lead in-class discussion | 10 pts |
| Referee report on JMP | 10 pts |
| Total | 100 pts |

Assignments are described in Section 3.

There is a degree of subjectivity in grading graduate coursework. I expect that all students who put an appropriate amount of effort into this course will receive an A or B. In the event that a student demonstrates a severe lack of effort, a lower grade may be appropriate.

### 2.4 Required course materials

The primary required text is Phaneuf & Requate (2017), *A Course in Environmental Economics: Theory, Policy, and Practice*, Cambridge University Press. All other readings will be provided electronically, on the course website or on Canvas. There are no other required purchases.

### 2.5 Additional criteria for successful completion

Students must complete all graded components (reading responses, research proposal, midterm, discussion lead, and referee report) to pass the course. Late assignments will not receive credit without a valid excuse.

### 2.6 Academic Integrity

Cheating is unacceptable. You are hereby reminded that you have pledged to uphold the honor code as follows:

> *Having read the Georgia Institute of Technology Academic Honor code, I understand and accept my responsibility as a member of the Georgia Tech community to uphold the Honor Code at all times. In addition, I understand my options for reporting honor violations as detailed in the code.*

Should you be caught cheating in this class you will be prosecuted according to the honor code and policies and procedures established by the Honor Advisory Council. Should you have any questions about this do not hesitate to contact me.

### 2.7 Student-Faculty Expectations Agreement

At Georgia Tech, it is important to strive for an atmosphere of mutual respect, acknowledgment, and responsibility between faculty members and the student body. See `http:`

//www.catalog.gatech.edu/rules/22/ for an articulation of some basic expectation that you can have of me and that I have of you. In the end, simple respect for knowledge, hard work, and cordial interactions will help build the environment we seek. Therefore, I encourage you to remain committed to the ideals of Georgia Tech while in this class.

### 2.8 Accommodations for students with disabilities

If you are a student with learning needs that require special accommodation, contact the Office of Disability Services at (404) 894-2563 or http://disabilityservices.gatech.edu/, as soon as possible, to make an appointment to discuss your special needs and to obtain an accommodations letter. Please also e-mail me as soon as possible to set up a time to discuss your learning needs.

## 3 Details on course assignments

#### Attendance and participation

Attendance is expected and will be necessary to succeed in the course. Participation in class discussions is strongly encouraged. Ask questions. If you do not understand something in class, you are almost certainly not the only one. If you never volunteer an incorrect answer, you aren't taking enough risks.
Attendance is not graded directly. Poor performance in this dimension may affect the final grade.

#### "Weekly" reading responses (20 points)

Students write a short response paper, roughly 250–300 words or two paragraphs, on designated readings. These readings are marked **[response due]** on the course website schedule. Expect eight to ten over the semester.
The response needs to go beyond summarizing the reading. Make connections to other topics in the course, to stories in the news, or to ongoing policy discussions. Reflect on how the reading updated your thinking. If you disagreed with a conclusion or a fundamental assumption, expand on that.
Focus on three questions:

1. What was one thing you enjoyed about the article?
2. What was one thing you did not understand, or would like to discuss in class?
3. What extensions or applications of the article would be interesting?

Submit via Canvas by 11:59 PM ET the day before the class in which the reading is assigned. That timing lets me read the responses before class. Late responses receive half credit.

#### Research paper proposal (35 points)

One goal of this course is to provide the tools necessary to conduct your own research in environmental, energy, and resource economics. Students develop a research proposal that introduces a novel and answerable research question, reviews related literature, discusses potential contributions, and begins to explore data sources and empirical methods. There is no explicit page requirement, though 10–15 pages with normal formatting is expected. The proposal should look and read like an extended introduction to an economics journal article: title, abstract, main body, background where relevant, conceptual framework, and references. Because this course focuses on economic theory applied to environmental problems, bring that conceptual thinking to your proposal even if the goal is an empirical paper.

Research takes time, with many fits and starts, so there are milestones through the semester. Start thinking about ideas as soon as possible and discuss them with your classmates and with me. Early feedback prevents rabbit holes. Topics may need revision if they are not aligned with the course or do not represent feasible projects. Ideally, proposals become third-year papers or dissertation chapters.

| **Milestone** | **Due** | **Points** |
|---|---|---|
| Project idea and research sketch | October 9 | 5 |
| Expanded outline and literature review | November 6 | 5 |
| Presentation | Last three classes | 5 |
| Final proposal | December 11 | 20 |

Students cannot use this proposal in conjunction with a writing assignment for another course. Proposals can sit at the intersection with other fields, for example environment and health, environment and development, or environment and trade.

#### Midterm (25 points)

In class, Class 15 (Monday, October 19).

#### Lead in-class discussion (10 points)

Each student leads or co-leads discussion of one paper in the topics section of the course.

#### Referee report on a job-market paper (10 points)

Students find a job-market paper by a current Ph.D. candidate working on an environmental, energy, or resource economics topic, and write a referee report. The goal is to see the end product of the Ph.D. on a topic of interest and to practice giving critical but constructive feedback.

I will provide guidance during the semester on where to search for job-market papers and on the structure of a referee report.

## Readings

The reading list is posted on the course website (`https://caseyjwichman.com/econ7102/`), organized by class meeting, with PDFs. The list is tentative and will change with the pacing of the course. The website is always the current version. Each week, I will also communicate where to focus your reading effort, for example on particular sections within chapters or papers that are assigned.

## Course Schedule

The class-by-class schedule—dates, topics, deadlines, slides, and the reading list—lives on the course website: `https://caseyjwichman.com/econ7102/`. Class numbers stay attached to their topic and slides, so a cancellation or delay changes which date a class falls on, not its number.